

Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 4A nr. 53.1

$$\begin{aligned} & \int \frac{dx}{\sin^4 x} \\ &= \int \frac{1}{\sin^2 x} \cdot \frac{1}{\sin^2 x} dx \\ &= \int \frac{1}{\sin^2 x} (1 + \cot^2 x) dx \\ & t = \cot x \left(dt = -\frac{1}{\sin^2 x} dx \right) \\ &= - \int (1 + t^2) dt \\ &= - \left(t + \frac{t^3}{3} \right) + C \\ &= -t - \frac{t^3}{3} + C \\ & \cot x = t \\ & \int \frac{dx}{\sin^4 x} = -\cot x - \frac{\cot^3 x}{3} + C \end{aligned}$$

References